

The principal methods of improving the heat resistance of niobium alloys are solid-solution alloying and strengthening with dispersed particles of refractory phase.

Solid-solution strengthening of refractory metals is most effective with mutual alloying with metals of groups V-VI, which form a continuous series of bcc solid solutions.

It has been found [1] that the heat resistance of alloys is intimately connected with the type of phase diagram at high temperatures, especially the position of the solidus line, i.e., the melting point. The melting point of niobium is raised most by the addition of tungsten, and lowered somewhat by tantalum and molybdenum (Fig. 1). All other metals of groups IV-V lower the melting point of niobium, leading to formation of a low point on the solidus-liquidus curve (Hf, Zr, V) or a system of the eutectic type characterized by a low solidus and a eutectic with a relatively low melting point (Cr, Mn, Fe, Co, Ni, etc.).

The change in the melting point with the addition of an alloying element can characterize the effectiveness of solid-solution alloying. The more the alloying element raises the melting point, the greater the extent to which it strengthens the metallic base at high temperatures, raising the heat resistance of the alloy. The effectiveness of solid-solution alloying can be determined from the difference in the melting points of the base metal and the respective solid solution (Fig. 1). Thus, to increase the high-temperature strength of niobium by solid-solution alloyings we recommend the addition of the most refractory bcc metals (W, Ta, Mo), with melting points higher than that of niobium. Metals forming low-temperature eutectics with niobium (Fe, Co, Ni, Cr, etc.), which are concentrated in the boundaries of grains and subgrains, sharply reduce the heat resistance and, particularly, the creep resistance [1].

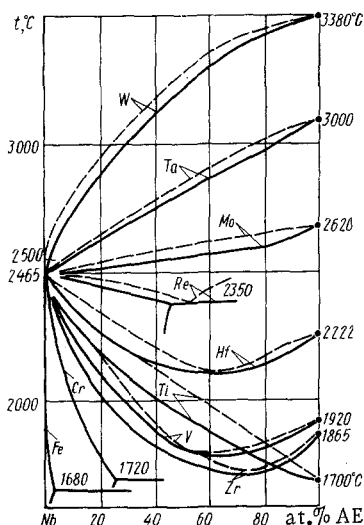


Fig. 1. Effect of alloying elements (AE) on the melting point of niobium. —) Solidus; - -) liquidus.

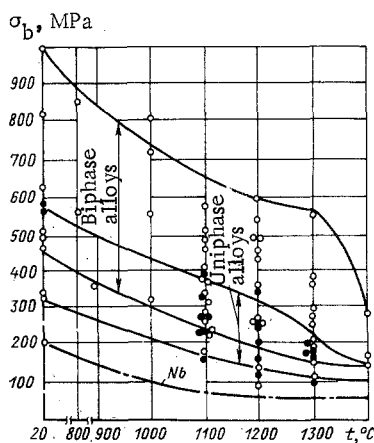


Fig. 2. Strength of single-phase and biphasic niobium alloys in relation to temperature.

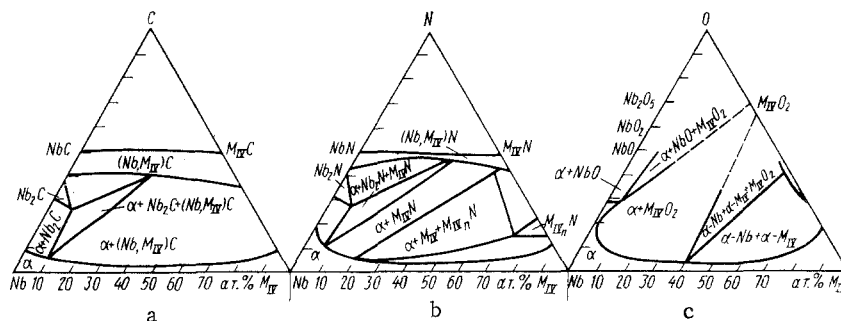


Fig. 3. Isothermal sections of ternary equilibrium diagrams [4] of Nb-MIV-C (a), Nb-MIV-N (b), and Nb-MIV-O (c). MIV is Zr or Hf. a) At 1800°; b) above 1200°; c) at 1700°.

At temperatures above $(0.45-0.55)T_m$ the effectiveness of solid-solution strengthening decreases. Thus, the increase in the strength of niobium alloyed with 20-30% Mo is retained up to 1300°C, while that of niobium alloyed with up to 30% W is retained up to 1400°.

Solid-solution alloying is widely used in industry to increase the heat resistance of niobium alloys, but strengthening with dispersed phases is increasing greatly (Fig. 2). Strengthening of refractory metals with intermetallic compounds is ineffective, since precipitation and coalescence occur at temperatures below operating temperatures. Metals are strengthened most effectively by refractory, thermodynamically stable compounds such as borides, carbides, nitrides, and oxides of transition metals. Thermodynamic analysis showed that the most promising compounds are those of titanium, zirconium, hafnium, and thorium. Biphasic alloys based on vanadium, niobium, chromium, molybdenum, and tungsten strengthened with carbides, nitrides, oxides, and borides of metals of groups IV and V are used at the present time.

Such alloys can be produced either by the metallurgical or P/M method as well as by internal oxidation or nitriding. The metallurgical method is the most widely used for preparing niobium alloys.

Refractory and strong borides, carbides, nitrides, and to some extent oxides dissociate and go into solution in melts of high-valency refractory metals of groups V and VI. Thus, regardless of the means of adding the alloying elements the final structure of the alloy will be determined by the conditions of crystallization of the melt and subsequent cooling of the ingot.

Phase equilibrium in these alloys is described by the equilibrium diagram of the $M_{V(VI)}-M_{IV(V)}-X$ system, where X is one of the interstitial elements (B, C, N, O). It follows

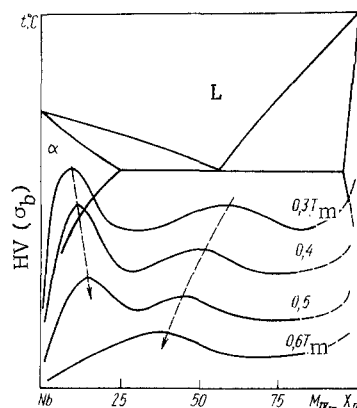


Fig. 4. Isotherms of the strength of alloys forming quasibinary eutectic cuts of $M_{IV}(VI)-M_{IV}X$. X is an interstitial element (B, C, N, O).

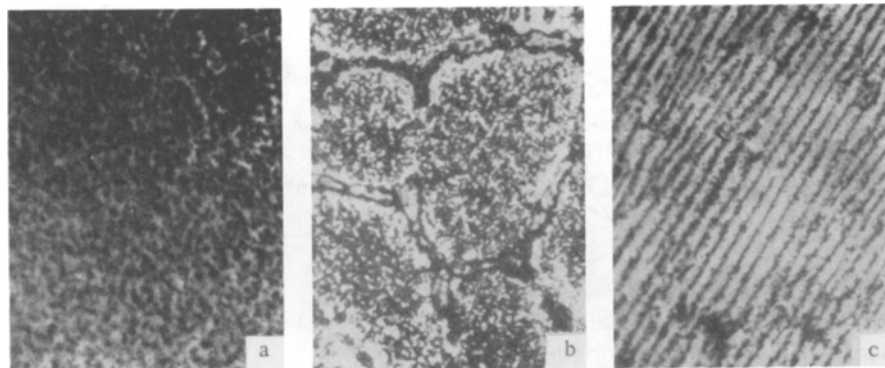


Fig. 5. Microstructures of Nb- M_{IV} -X alloys. a) Precipitation-hardening alloy, 75,000 $\times$; b) dispersion hardened alloy, 1000 $\times$; c) eutectic (directionally solidified), 440 $\times$.

from an analysis of isothermal and polythermal sections of ternary diagrams of Nb- M_{IV} -C(N, O, B) (Fig. 3) that many of these systems are characterized by a quasibinary eutectic cut. The solid solution based on niobium (α -Nb) is in equilibrium both with compounds of the Nb_2C , Nb_2N , and NbO type and with compounds of the alloying metal such as $(Nb, Zr)C$, ZrN , and ZrO_2 (Fig. 3). Strengthening is greatest for the alloys in regions where α -Nb and the corresponding compound $M_{IV}X$ are in equilibrium. For the Nb-Zr(Hf)-O(N) system these regions correspond to quasibinary cuts of the eutectic type. Among the metals of group IV, titanium is less promising for niobium alloys, since the thermodynamic strength of titanium carbides, nitrides, and oxides is inferior to that of the corresponding compounds of zirconium and hafnium.

According to the accepted principles [1] of the variation of strength with the composition of alloys whose equilibrium diagrams are eutectic, there should be two peaks. The first strength peak should be found in alloys with elements in amounts close to the limits of variable solubility of $M_{IV}X$ in $M_{IV}(VI)$ (precipitation-hardening alloys), and the second in alloys whose composition is close to the eutectic (Fig. 4). For example, for the Nb-ZrN and Nb-HfN systems the first peak corresponds to niobium alloys with 2-3.5% N (by weight) at a ratio of at.% M_{IV} /at.% N $\cong$ 1 [3, 4, 6]. At temperatures above (0.4-0.5) T_m this peak declines and disappears due to the coalescence of dispersed particles, which leads to suppression of the precipitation hardening effect. For aging alloys with only slight solubility of $M_{IV}X$ in $M_{IV}(VI)$ no peak strength is manifest even at low temperatures, which is characteristic of V-ZrN and Mo-TiC alloys [3, 7].

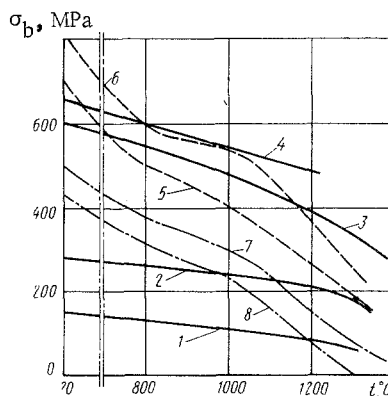


Fig. 6. Temperature dependence of the ultimate tensile strength of niobium alloys hardened with carbides (solid lines), nitrides (dashed lines), and oxides (dash-dot lines). 1, 2, 3, 4) 1, 2.5, 5, and 9 mole % $M_{IV}C$, respectively; 5, 6) 1 and 3 mole % $M_{IV}N$, respectively; 7, 8) 1 and 3 mole % $M_{IV}O_2$, respectively.

The peak strength corresponding to alloys with a composition close to the eutectic of the M_1 - M_2 system decreases when the temperature is raised to $(0.4-0.5) T_m^{eu}$ and at temperatures above $0.6T_m^{eu}$ gradually changes to a low point [1]. For alloys of the $M_V(VI)$ - $M_{IV}C(N, B, O)$ system the lowering of the peak strength is accompanied by a shift to a higher concentration of the metallic component on the diagram because of coalescence of particles of hardening phase. At a temperature above $0.6T_m$ the minimum corresponds to the eutectic composition, while the highest strength shifts into the region of hypoeutectic compositions (Fig. 4).

It is known that with a great difference in melting points of the original components a so-called degenerate or differentiated eutectic [8] may be formed in eutectic alloys in the process of crystallization with little supercooling. The phases entering into such a eutectic are in the form of large separate crystals. In $M_V(VI)$ - $M_{IV}X$ systems, e.g., a degenerate eutectic with individual precipitates of strengthening compound $M_{IV}X$ is formed (Fig. 5b).

With rising temperatures the coalescence of the strengthening particles speeds up; the larger the amount of compound $M_{IV}X$ in the alloy, the larger these particles become. These processes also induce a shift of the peak hardness and strength to the hypoeutectic region. The extent of the shift varies with the specific system. For example, at high temperatures (above $0.5T_m$) the strongest alloys are not the precipitation-hardening (aging) or eutectic alloys but the corresponding hypoeutectic alloys. The increase in the strength of these alloys is determined by the large volume of strengthening phase.

On the basis of the physicochemical concepts presented and with use of the $M_V(VI)$ - $M_{IV}X$ phase diagram it is possible to select the optimal composition of alloys hardened by dispersed phases for specific operating conditions.

This approach was verified by studies of the Nb-Hf(Zr)-C(N,O) system. The effectiveness of precipitation hardening of aging alloys depends mainly on the solubility of the $M_{IV}X$ compound in the metallic base, which determines the quantity of hardening phase precipitating from the supersaturated solid solution. Because of the much higher solubility of nitrogen [9] and nitrides ZrN (HfN) [10] than oxygen [9] and oxides ZrO_2 (HfO_2) [16] in niobium, the region of the primary solid solution based on niobium on the phase diagrams of the Nb- $M_{IV}N$ system (Fig. 3) is wider than on diagrams of the Nb- $M_{IV}O$ system. Other conditions being equal, the effectiveness of hardening with refractory nitrides is larger than for hardening with refractory oxides for alloys of this group (Fig. 6).

Previous studies of the rules governing the formation and decomposition of solid solutions in Nb-Zr(Hf)-O(N) alloys showed that processes typical of classical aging alloys occur

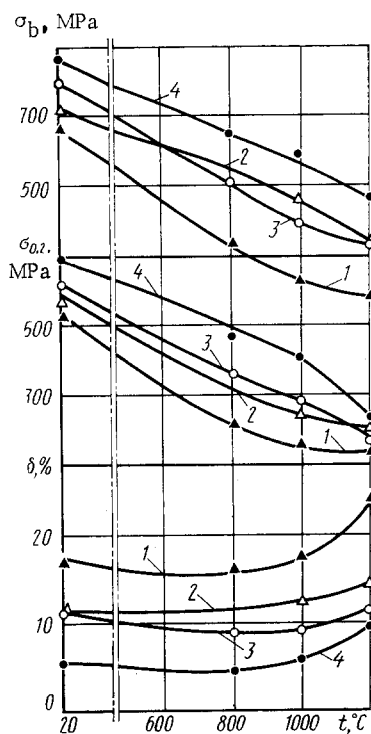


Fig. 7. Effect of testing temperature on the ultimate tensile strength of niobium alloyed with 2 mole % ZrN after different heat treatments. 1) Annealing at 1400°C for 1 h; 2) annealing at 1800° for 1 h; 3) quenching in a stream of helium from 1700° (1 h) + aging at 1000° for 1 h; 4) quenching in a stream of helium from 1700° (1 h) + aging at 1000° for 20 h.

in them [11]. For this reason, high strength of precipitation-hardening alloys of this system is achieved by heat treatment (Fig. 7).

On the diagram of the Nb-Zr(Hf)-C system, to which carbide-hardened alloys belong, a triphase region [α -Nb-Nb₂C-(Nb, M_{IV})C] is visible near niobium (Fig. 3), which prevents formation of a quasibinary eutectic between niobium and ZrC or HfC. In alloys with compositions near Nb-M_{IV}C there is also a triphase region with an initial structure determined by decomposition of the supersaturated solid solution in the binary Nb-C system with precipitation of metastable carbides Nb₂C and Nb₃C₂ [4]. These carbides are precipitated in alloys containing up to 2-3 at.% Zr, Hf, and C (each). Due to the lower solubility of carbon in niobium as the temperature is lowered [9] and also because of the high diffusion mobility of carbon even during cooling from quenching temperature, the solid solution of carbon in niobium decomposes rapidly. In the process of holding a niobium alloy of the Nb-C system at 800° coarse particles of Nb₂C, which do not ensure effective hardening [12], are precipitated within 4 min. Thus, precipitation-hardening alloys with a structure determined by decomposition of the α -solid solution in the binary Nb-C system are not very promising for operation at temperatures above 1000°.

Under certain conditions metastable Nb₂C transforms to more stable (Nb, M_{IV})C [3, 4]. The considerable mutual solubility of NbC and ZrC (or HfC) [4, 13] determines the formation of complex carbide (Nb, Zr)C or (Nb, Hf)C in ternary alloys. However, even after transformation of Nb₂C into (Nb, Zr)C, which occurs in carbide-hardened alloys with rising aging temperatures or increasing time, the precipitation hardening is not as effective as in precipitation-hardening alloys with nitrides and oxides (see Fig. 6).

The most widely used carbide-hardened niobium alloys are those containing not over 1-2 at.% Zr (Hf) and C (each) and alloyed with molybdenum and tungsten. The preferential use of alloys with carbide hardening is evidently due to the simple technology of melting and good workability.

In analyzing the structure and properties of alloys with 1-2 mole % ZrC it is necessary to keep in mind the presence of oxygen that enters in the process of melting and subsequent conversion. Zirconium or hafnium dissolved in niobium and not participating in the formation of metastable niobium carbides interacts with oxygen, forming ZrO₂ or HfO₂. These dispersed oxides provide hardening that sometimes exceeds hardening by particles of (Nb, M_{IV})C (see Fig. 6).

The possibility of obtaining precipitation-hardening alloys is determined by the width of the hypoeutectic region on the quasibinary eutectic cut. According to [5], the limit

TABLE 1

System	Amount of ZrO ₂ (HfO ₂) in eutectic			Max. solubility of ZrO ₂ (HfO ₂) in Nb, mole %
	mole %	wt. %	vol. %	
Nb—ZrO ₂	1.14	1.5	2.6	0.015
Nb—HfO ₂	0.9	2.0	1.5	0.015

solubility of the oxide in the metal on cuts of Nb—ZrO₂ and Nb—HfO₂ is near the eutectic composition (see Table 1). Even a slight increase in the amount of ZrO₂ (or HfO₂) above the maximal solubility leads to a large quantity of coarse hypereutectic oxides in the structure. This alloy is of little promise from the viewpoint of heat resistance. The ultimate tensile strength of alloys with 3-5 mole % ZrO₂ at 1200° is 50-70 MPa, while $\sigma_b = 140-200$ MPa at 1200° for precipitation-hardening alloys containing less than 1 mole % ZrO₂.

In the quasibinary Nb—ZrN and Nb—HfN systems the region of hypoeutectic alloys is fairly wide [10], thus permitting the use of precipitation hardening. The most interesting precipitation-hardening alloys belong to the Nb—Zr(Hf)—C system. Dual equilibrium between the solid solution and carbide (Nb, Zr)C is established at a concentration of more than 2 at.% Zr and C (each). Since this portion of the cut is characterized by the presence of eutectic corresponding to 12 mole % ZrC, the cut can be regarded as quasieutectic with a fairly broad region of hypoeutectic alloys. A similar approach can be used for Nb—Hf—C alloys.

Tests of the mechanical properties of precipitation-hardening Nb—Zr—C alloys showed that they are characterized by high strength and relative strength at temperatures of 1000° and higher and quite stable mechanical properties at elevated temperatures (Fig. 6). Alloys with a high volume percentage of carbide phase, in contrast to niobium and its single-phase alloys, are susceptible to intensive strain hardening, which in combination with fairly high recrystallization temperatures (1500° and higher) additionally increases the heat resistance.

Application of alloys with a large concentration of carbide phase are limited, however, by their low ductility at low temperatures. This results from the structural microheterogeneity that is characteristic of a cast material, a supersaturated solid solution, the presence of thermal stresses and stresses that occur during decomposition of the solid solution as the ingot cools, and, mainly, from precipitation of a large quantity of eutectic carbides with an unfavorable morphology from the standpoint of ductility (Fig. 5b).

To utilize these alloys it is necessary to improve the cast structure, which can be attained by proper heat treatment, deformation by hydrostatic extrusion [14], modification [3], and other methods. Recent developments in powder metallurgy offer broad possibilities of obtaining a homogeneous structure with dispersed and evenly distributed carbide phase in alloys with a large volume percentage of carbides.

By selecting the composition of Nb—Zr(Hf)—C alloys with a large volume percentage of carbide phase to ensure precipitation hardening at high temperatures and by selection of the proper heat treatment for wrought alloys ensuring strain hardening by optimizing the structure it is possible to increase the relative strength at 1200° to 5-6 Km, which is considerably higher than the relative strength of commercial niobium alloys.

In recent years attention has shifted to eutectic alloys of refractory metals with refractory compounds such as the binary eutectic alloy Nb—Nb₂C [15], which is a natural composite material with a relative strength exceeding that of most commercial niobium alloys. The production of alloys with reinforcing filaments of more refractory carbides permits the hardening effect to be increased still more, especially with use of directional solidification (see Fig. 5c).

Thus, we still have a fairly large reserve for increasing the heat resistance of niobium alloys, especially the combination of solid-solution alloying with refractory metals that have a melting point higher than that of niobium and hardening with dispersed phases such as oxides, carbides, and nitrides of transition metals of group IV. From the physico-

chemical viewpoint, the heat resistance is promising for three groups of alloys from the biphasic alloys of this type: 1) precipitation-hardening (aging); 2) dispersion hardened; 3) eutectic.

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